

Name: _____ **Date:** _____

Definition 1. Recall the *chain rule* (I don't have a mnemonic for this one):

$$\frac{d}{dx} \left(f(g(x)) \right) = f'(g(x)) \cdot g'(x) \quad \text{or} \quad (f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

Problem 1. For each of the following functions, use the chain rule to find the derivative $\frac{dy}{dx}$. Hint: first, figure out what the “inner” and “outer” functions are! You may also have to the other derivative rules too!

a) $y = e^{3x}$

b) $y = e^{x^2}$

c) $y = e^{\sin(x)}$

d) $y = \sin(\cos(x))$

e) $y = \sin(3x^2 + 2x - 1)$

f) $y = \tan(e^x)$

g) $y = \tan(\ln(x))$

h) $y = \sin^2(x)$

$$i) \ y = e^{x \sin(x)}$$

$$j) \ y = \ln(\sin(3x))$$

$$k) \ y = \tan(\sqrt{x})$$

$$\ell) \ y = \sqrt{\sin(x)}$$

$$m) \ y = \frac{\sin(x)}{e^{\cos(x)}}$$

$$n) \ y = \tan\left(\frac{x}{e^x}\right)$$

$$o) \ y = \sin(e^{\sin(x)})$$

$$p) \ y = \sqrt{\ln\left(\tan(x^3 - x + 12)\right)}$$